Homework One

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1. (5) Define a lattice.

2. (5) How many lattice points are associated with (a) a primitive cell? (b) a unit cell?

3. (5) The atoms associated with a lattice point are referred to as the ___________________?

4. (15) For the fcc lattice, the basis vectors of the cubic unit cell are *a* times the unit vectors in the x, y, and z directions.
(**a**) What are the the *primitive* vectors ($\boldsymbol{a}$, $\boldsymbol{b}$, and $\boldsymbol{c}$) corresponding to the primitive cell? Hint: They are non-orthogonal. They are all the same length. They are shorter than *a*. An integer multiple of them (n$\boldsymbol{a}$ + m$\boldsymbol{b}$ + p$\boldsymbol{c}$) will reach every lattice point in the fcc lattice (and no other points). Hint: Consider the nearest neighbor lattice points.
(**b**) What is the ratio of the volume of the primitive cell to the volume of the cubic unit cell? You find the volume from the magnitude of the triple product $V = |(\boldsymbol{a} \times \boldsymbol{b}) \cdot \boldsymbol{c}|$.
(**c**) Does this ratio make sense in terms of the number of lattice points contained in the primitive cell versus the unit cell? Explain.

5. (10) Beginning with an fcc lattice, adding atoms at $\frac{a}{4}(1,1,1)$ from each fcc atom gives the diamond lattice as shown on slide 20 of the Lec. 2 slides.
Give the position (i.e. the x, y, z coordinates) of each one of the 4 interior atoms. The first one is already given as $\frac{a}{4}(1,1,1)$, i.e. the back left corner atom is assumed to be at the origin (0,0,0) with the cube going out along the positive x, y, and z axes.

6. (20) Find the number of atoms/unit-cell and nearest neighbor distance, in terms of the edge length *a*, for (a) sc, (b) bcc, (c) fcc, (d) diamond, and (e) zinc blende unit-cells. Assume that for (a-c) there is one atom per lattice point. Note that the nearest-neighbor distance corresponds to the atomic bond length.

7. (20) Calculate the density (in units of g/cm^3) of GaAs from the lattice constants, atomic weights, and Avogadro's number. The atomic weights of Ga and As are 69.7, and 74.9, respectively. The lattice constant corresponding to the cubic unit cell of GaAs is 5.65 (angstroms), respectively.

8. (10) Which of the following unit cells are primitive cells for the 2-dimensional lattice? Explain.